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Lead Software Engineer · Agent Harnesses · Full-Stack TypeScript · AWS

WHAT I BUILD

Two kinds of work, and they turn out to be the same job. **Live media pipelines** that run unattended on events nobody gets to re-run: a headless browser capturing the show, ffmpeg encoding it out, and a reliability layer that assumes the capture will die mid-session. I've run that pipeline as a managed service on AWS Elemental MediaLive and as a self-hosted ffmpeg stack on ECS Fargate, across the migration between them.

The other kind is the **harness around a model**: the execution loop, the tool surface, the verifier, and the cost controls. Most of it runs on Claude. Both are long-running unattended processes where the thing reporting success is the thing you can least afford to believe.

When the human checkpoint comes out, an automated judge has to say whether the action was right. Most of my design time goes into making that judge catch the runs that look right and are not. The rest of the harness is reproducibility, so a run can be debugged afterwards, and supervision, so a person can stop one mid-flight while trust is still being earned. Per-call cost accounting keeps the bill survivable at volume. Thirteen years shipping and operating production software underneath all of it.

TECHNICAL SKILLS

Agent Tooling:	MCP Protocol, Anthropic Claude API, OpenAI API, Tool/function-call schema design, Agent orchestration, LLM-as-judge evaluation, Deterministic replay, Token-budget & cost accounting, Playwright / CDP
Languages:	TypeScript (6 yrs, primary), Go, Rust, Dart, Python (scripting & test tooling)
Backend:	Node.js, GraphQL, tRPC, Prisma, PostgreSQL, MongoDB, Redis, BullMQ, Serverless
Media & Capture:	ffmpeg, HLS / LL-HLS, RTMP ingest & fan-out, H.264 ABR ladders, Headless capture of live WebRTC (Puppeteer / CDP), AWS Elemental MediaLive, MediaPackage, Amazon Chime SDK, Mux
AWS:	ECS / Fargate, Lambda, CDK, S3, SQS, EFS, CloudFront, CloudWatch
Frontend & Mobile:	React, React Native, Flutter / Dart, Vue 3, Next.js
Infrastructure:	Kubernetes, Docker, GitHub Actions, k6 load testing

AGENT HARNESSES & EVALUATION

browxai—MCP-native browser control for agents (private) 2026

Stack: TypeScript, MCP, Playwright, CDP, Node.js

- Built a model-agnostic browser-control server exposing a token-efficient Playwright/CDP surface over MCP. The consumer is a model on a context budget, so I designed for fewer, better-shaped operations with predictable failures.
- Session isolation, trusted-input dispatch, network and console capture, structured snapshots of the accessibility tree, typed extraction. An agent needs all of it to act on a page and then prove what happened.
- Serves as the execution substrate for wrightxai and docsxai.

wrightxai—SWE-style web agent with a judge gate (private) 2026

Stack: TypeScript, Claude API, code-as-action, LLM-as-judge

- Implemented a single-LLM loop with code-as-action execution, in the shape of Microsoft's Webwright: the agent writes and runs code to act.
- Designed the **screenshot judge gate**. The agent's own claim of success is treated as unreliable. A separate judged pass over the resulting visual state decides whether the task completed. A lenient judge trains behaviour that games the measurement, so the work went into rejecting confident, well-formed, wrong trajectories.

docsxai—documentation agent with deterministic replay (private) 2026

Stack: TypeScript, Claude Code plugin, MCP

- Built an LLM-agnostic engine that walks a running web application and emits screenshot-rich user documentation, shipped as a Claude Code plugin.
- **Deterministic replay** was a hard requirement, so the pipeline is built for the same walk to emit the same artefact every run.

composerxai—a communication framework for agentic work 2026

Stack: TypeScript, Node.js, Claude Code plugin, zero runtime dependencies

- Open-sourced a framework for how an agent decides what to write, and how it reports back to the person running it. Ships as a plugin with two skills, six register definitions, five document templates and a prose linter carrying 22 rules.
- Every rule is classified enforceable or judgement, and the tool prints which. Enforceable rules gate the exit code. Judgement rules report and never fail a build.
- The rules come from measurement. Structural tells survive adversarial prompting, so the linter weights structure above word lists and stamps every lexical entry with the era it was measured in. An earlier attempt to build a *detector* scored below a baseline of three regexes and a standard deviation; that negative result is published in the repository, along with the ~180 sources behind the rules.
- The reporting side is calibrated too. Probability language is a contiguous partition, with the numeric band stated inline every time a term is used. Escalation runs on three tiers, separating what stops work from what is only signalled.

Context Specification Language—Rust reference toolchain (private) 2026

Stack: Rust, XML parsing, eval harness, CLI

- Wrote parser, validator, normaliser, CLI and repository-aware file-reference checking for a context specification language.
- Built the eval harness alongside it, so a change to the spec is scored against a fixture corpus.

remotxai—supervision for long-running agent sessions (private) 2026

Stack: TypeScript, encrypted relay, WebSockets, PWA

- Built a remote supervision platform for long-running AI coding-agent sessions: encrypted relay server, real-time terminal streaming, mobile-first PWA. A person can watch an agent that has been running for hours, interrupt it and steer it from a phone.
- Aimed at agents that act on the world, where oversight has to land while the run is still going. The supervision channel is live, so a person intervenes mid-run.

EXPERIENCE

Founder / Principal Engineer · Kalebtec Mar 2026 – Present

Stack: TypeScript, Rust, MCP, Claude API, Anthropic SDK, Playwright, React, Vue 3, Node.js, AWS

- Work is Claude-first day to day: designing prompts and tool schemas, orchestrating multi-step tool-calling, building the review passes that catch what AI-generated code gets wrong, and tracking spend per call so cost is measured.
- Delivered **Bardi Fuels**, a private market-intelligence platform for oil/gas and biofuels professionals with real-time prices, bids/offers, newsletters and trade recaps; shipped to iOS via Fastlane and the App Store.

Lead Software Engineer · Live Platform · FanFest Nov 2024 – Present

Stack: OpenAI, Vue 3, TypeScript, GraphQL, Prisma, AWS (ECS/Fargate, Lambda, CDK), ffmpeg, Stripe, Ably, Socket.io

- Built and owned the live-broadcast pipeline end to end, from AWS Elemental MediaLive through its migration to a self-hosted **ffmpeg** encoding stack on ECS Fargate: headless-browser capture, an H.264 ABR ladder, HLS packaging, and RTMP fan-out to YouTube, Facebook, X and TikTok from a single encode.
- Built the studio-production ingest path: a produced show, run for partners including AC Milan, streams into the broadcast over **RTMP through Mux**, so a live gallery feeds the platform directly and the mass audience scales off that same feed.
- **Sole-authored its reliability layer:** a validated lifecycle state machine, plus 10-second heartbeats reporting ffmpeg, browser and socket liveness as separate signals. An origin-reachability probe covers the case where the heartbeat is fresh and the output is already dead.
- Made a dropped capture recoverable. A watchdog replaces the failed instance under optimistic locking, with bounded attempts and a cooldown. Resume-from-object-storage stitches the replacement into one continuous recording, so a show that drops at minute twenty still produces one usable asset.
- Built an object-storage circuit breaker that suppresses replacement launches when the store is itself the fault, a bounded 90-second shutdown flush that checkpoints for post-processing, disk-pressure cleanup that never touches the live edge, and a no-restart crash policy, so a container out of memory or disk is replaced outright.
- Architected **Simulfest**, the interactive broadcast format: a headless browser joins and captures a live **Amazon Chime** WebRTC conference, and that session is simulcast to the mass audience through MediaPackage multi-resolution HLS over CloudFront, reaching tens of thousands of viewers per event.
- Own the moderation and translation pipelines on a fan-engagement platform serving PSG, Manchester City, AC Milan, the 49ers and OG Sports, moderating user-generated messages live during events, under broadcast load.
- Built the multilingual pipeline: EventBridge to SQS FIFO grouped by object, eight target languages returned from a single model call, per-language failure isolation so one failed language re-enqueues alone, and content-hash idempotent upserts keyed on object and language so a retry is never a duplicate.
- Enforced structural invariants through the prompt, so a round trip preserves interpolation placeholders, emoji, URLs and markdown. Malformed returns hit a JSON-repair fallback.
- Built the real-time platform underneath it: live chat, loyalty queues and Stripe membership flows, with k6/ECS load-testing validated to **20,000 virtual users** on that path (the video path is load-tested separately).

Lead Software Engineer · v3 Rebuild · FanFest

Nov 2023 – Nov 2024

Stack: Vue 3, TypeScript, GraphQL, Prisma, AWS, pnpm monorepo, Socket.io, OpenAI

- Led a team of 6 engineers architecting FanFest v3, a Vue 3/TypeScript/GraphQL/Prisma/AWS pnpm monorepo replacing a split-repo Vue 2/Parse Server/GCP stack.
- Delivered architecture for live chat, memberships, loyalty, SDK embeds and AI-enabled moderation; established shared TypeScript packages across frontend, API and SDK apps so contracts are enforced by types.

Senior Frontend Engineer · WSC Sports (Clipro) (Contractor)

Aug 2025 – Jun 2026

Stack: React, TypeScript, Vite, Redux, Jest, Playwright

- Sole frontend engineer on the AI voiceover squad for close to a year, owning the domain end to end (Clipro is used by the NBA, ESPN and 300+ broadcasters and rights holders).
- Re-architected the AI voiceover frontend to an async, job-based model, the right shape for long-running generative work: progress, partial failure and retries all became states the user can see.
- Ran frontend technical reviews and planning across the squad; onboarded and mentored the incoming engineer and grew automated coverage with Jest and Playwright.

Senior Full-Stack Engineer · Limbic AI (Contractor)

Oct 2023 – Feb 2024

Stack: Next.js, React, TypeScript, Node.js, PostgreSQL, MongoDB, BullMQ, AWS

- Built structured chatbot dialogue flows and rule-based risk detection for a UK mental-health screening platform, under clinical safety constraints. The detection is rule-based by design, so every escalation can be explained to a clinician line by line.
- Built a queued emergency-notification system with retries and time-travelling dialogue state, so an escalation still reaches a human when a downstream service is failing.

Mobile Engineer—Identity & Digital Wallets · Ping Identity (Contractor) Mar 2024 – Jul 2024

Stack: Flutter, Dart, Swift, Java, verifiable credentials, DID, BLE, QR

- Built a cross-platform proof-of-concept in digital identity—verifiable-credential issuance, storage, verification and presentation across iOS and Android, with native platform channels bridging Swift and Java SDKs.

Senior Software Engineer · GraphAware (Hume) (Contractor) May 2023 – Sep 2023

Stack: Vue 2, TypeScript, Java, Spring, Neo4j, Playwright, Cypress, Testcontainers

- Delivered features for an enterprise knowledge-graph platform covering graph exploration, geospatial analysis, schema management and orchestration workflows.
- Wrote and maintained E2E suites in Java Playwright and Cypress covering complex canvas-based graph-editor interactions.

Lead Software Engineer · Synphonyte May 2020 – Jan 2024

Stack: TypeScript, Node.js, Go, Rust/WASM, Flutter, Dart, Vue, Nuxt, React, Next.js, PostgreSQL, Kubernetes, Three.js

- Led development across multiple concurrent client products over 3.5 years, managing a team of up to 7 engineers.
- Built **Messynger**, a multi-channel conversation platform unifying WhatsApp, Messenger, Slack and MS Teams into one inbox on IBM Cloud. Each channel arrives with its own threading, identity and delivery semantics, and the platform normalises all four into one conversation model so a team works every thread from a single surface.
- Delivered Voxel, a browser-based dental 3D imaging platform with Three.js/WebGL2, Rust/WASM geometry, volumetric scan interaction, and pixelmatch render-diff automation.
- Delivered Sunliner, a travel booking portal for a Dutch holiday-park operator covering package search, availability calendars, booking flows and an admin dashboard, with Go on the service side.
- Built a Deno/Puppeteer scraping pipeline for hotel data ingestion, and published a standalone LitElement web component to npm.

Full-Stack & Mobile Engineer · Learapido May 2018 – Feb 2020

Stack: Flutter, Dart, React Native, Expo, Ruby on Rails, Node.js, Express, GraphQL, Firebase, MongoDB

- Owned two generations of the mobile app and led the move from React Native/Expo to Flutter, which became the primary client: onboarding, auth, activities, diagnostics, Firebase dynamic links and in-app purchases.
- Built the Rails API behind it, covering training routines, diagnostics, subscriptions, activation codes and user management.

Mobile & Frontend Engineer → Frontend Lead · Humantelligence Oct 2018 – Nov 2023

Stack: React Native, React, TypeScript, Vite, Turborepo, Zoom Apps SDK, Webex Embedded Apps, Microsoft Teams / TeamsFX, Ruby on Rails, MongoDB

- Shipped in-meeting app surfaces for Zoom, Webex and Microsoft Teams from a single monorepo, with per-app Vite build and S3/CDN publish flows.
- Delivered the consumer mobile app (iOS/Android) in React Native from inception and built the talent-analytics web platform, including a Chrome extension embedded in Gmail and Google Calendar.
- Led the web frontend modernisation: migrated a legacy ejected-CRA/Webpack codebase to TypeScript + Vite + a Turborepo/pnpm monorepo, delivering a 10x improvement in development and deployment performance (client-reported).

EDUCATION & CERTIFICATIONS

Self-taught, 13 years of continuous professional practice since 2013, programming since 2010.

Certifications: Apollo Graph Developer Professional (2023) · Tigera Calico Kubernetes Networking & Security (2023) · Argo CD GitOps (2023) · BerkeleyX Bitcoin & Cryptocurrencies (2022) · Observe Practical Observability (2023)